Agent Failure Labs
Product Overview for University Pilots

Hands-On AI Agent Security Training with Zero Infrastructure Burden

Executive Summary

Agent Failure Labs is a hosted educational cyber range for teaching AI agent security. Students attack intentionally vulnerable AI agents, inspect trace-backed evidence, explain the root cause, and submit structured security reports. Instructors receive ready-to-use teaching materials, grading rubrics, sample reports, and hosted lab access without needing to deploy local infrastructure.

1. The Problem

University cybersecurity courses are under pressure to teach emerging AI security risks, but most instructors face practical barriers.

- AI agent security is a fast-moving topic.
- Real systems are unsafe and inappropriate for student experimentation.
- Toy prompt examples often fail to show realistic tool use, traces, or operational impact.
- Local environment setup creates avoidable friction for students and teaching staff.
- Instructors need assessment evidence, not just interactive demos.
- Curriculum reviewers increasingly expect clear alignment with cybersecurity and AI risk frameworks.

The result is that many courses can describe AI agent risks in lectures, but struggle to provide safe, hands-on, assessable practice.

2. The Agent Failure Solution

Agent Failure Labs provides browser-based, hosted AI agent security labs for university teaching.

Students work inside controlled scenarios where vulnerable AI agents interact with tools, external content, synthetic sensitive data, and trace logging. The labs are designed so that students can observe both the exploit outcome and the system-level failure that caused it.

Core Value Proposition

Agent Failure gives instructors a ready-to-assign AI security lab module: empirical, hands-on, trace-backed, standards-aligned, and hosted without requiring faculty to manage infrastructure.

3. What Students Do

In a typical Agent Failure lab, students:

1. read a mission briefing;
2. interact with a vulnerable AI agent;
3. use a controlled attacker interface;
4. trigger a realistic AI security failure;
5. inspect event timelines and traces;
6. collect evidence;
7. explain the root cause;
8. propose mitigations;
9. submit a structured security report.

This shifts the learning experience from passive awareness to active security analysis.

4. Flagship Pilot Lab

The initial pilot centers on Lab 1.

Field	**Description**
Lab title	OpsMail Assistant: Indirect Prompt Injection via Poisoned Inbox
Primary topic	Indirect prompt injection in AI agents
Student role	Security tester in a controlled educational cyber range
Target system	AI email assistant used for onboarding operations
Attack vector	Attacker-controlled email content
Target asset	Synthetic manager home address
Core failure	Untrusted email content influences the assistant's behavior
Student output	Trace-backed security report with root-cause analysis and mitigations
Estimated lab time	15–25 minutes hands-on; 60–90 minutes including briefing and report discussion

5. Why This Lab Matters

Indirect prompt injection is an important AI agent security failure mode. The attacker does not need to type directly into the victim's chat interface. Instead, the attacker places malicious instructions inside content that the agent later reads, such as an email, document, webpage, ticket, or knowledge-base entry.

This matters because modern AI agents often combine:

- instruction following;
- external content retrieval;
- tool use;
- access to sensitive data;
- autonomous or semi-autonomous workflows.

When these features are combined without strong trust boundaries, untrusted content can influence agent behavior and cause real security consequences.

6. What Instructors Receive

The pilot pack is designed to reduce instructor preparation time and adoption risk.

Artifact	**Purpose**
Product overview	Explains the value proposition and pilot model.
Syllabus mapping	Shows where the lab fits in a 15-week course.
Student handout	Gives students the background, rules, objective, and deliverables.
Instructor guide	Provides solution key, troubleshooting guidance, and failure-mode analysis.
Slides	Supports a short theory briefing before the lab.
Rubric	Enables objective, constraint-based grading.
Sample student report	Shows what an excellent submission looks like.
Standards coverage matrix	Maps the lab to cybersecurity and AI risk frameworks.
Technical access guide	Explains hosted access, browser requirements, and network needs.
Privacy and security FAQ	Addresses data handling, trace logs, runtime isolation, and institutional concerns.
Pilot feedback form	Measures instructor experience, student engagement, and adoption friction.

7. Zero Infrastructure Management

Agent Failure is designed for hosted university pilots.

The goal is to let instructors assign an AI agent security lab without becoming platform operators.

8. Educational Differentiators

8.1 Empirical, Not Just Conceptual

Students do not merely read about prompt injection. They perform an attack in a controlled environment and observe the resulting failure.

Instructor or Student Requirement	Pilot Model
Local Docker setup	Not required
Local Python or Node.js installation	Not required
Student LLM API keys	Not required
University server deployment	Not required for standard pilot use
GPU access	Not required from the institution
Command-line tooling	Not required
Browser access	Required
Internet connection	Required
Course link, class code, or login	Required depending on pilot configuration

8.2 Trace-Backed Evidence

The platform records relevant events such as prompts, tool calls, tool outputs, assistant responses, and success conditions. Students use these traces to support their reports.

8.3 Assessment-Oriented

The lab is designed around reportable evidence, not only interactive exploration. Instructors can grade using a structured rubric.

8.4 Safe Synthetic Environment

The lab uses fictitious people, fictitious organizations, and synthetic sensitive data. Students are not attacking real users or production systems.

8.5 Standards-Aware

The lab supports selected learning outcomes related to secure design, privacy, prompt injection, threat modeling, trace analysis, and AI risk management.

9. Instructor Benefits

- Reduces preparation time for teaching AI agent security.
- Avoids local environment setup problems.
- Provides a polished hands-on lab without requiring infrastructure management.
- Gives students a realistic but controlled security scenario.
- Produces trace-backed evidence for grading and feedback.
- Supports syllabus and standards alignment.
- Enables pilots with small groups before broader course adoption.

10. Student Benefits

- Hands-on practice with an emerging AI security failure mode.
- Experience using traces and event logs as security evidence.
- Practice writing a professional vulnerability report.
- Exposure to agent-specific risks involving tools, external content, and sensitive data.
- Clear rules of engagement inside a safe educational environment.

11. Recommended Pilot Format

Pilot Element	Recommended Configuration
Pilot scope	One lab: OpsMail Assistant indirect prompt injection
Class size	20–80 students, or a smaller review group for first trial
Student time	60–90 minutes including briefing, lab, trace review, and report setup
Instructor preparation	20–40 minutes using the pilot pack
Assessment	Short security report using the provided rubric
Delivery mode	In class, remote, workshop, or homework lab
Technical model	Hosted browser-based platform
Feedback collection	Instructor and student feedback after completion

12. What Success Looks Like

A successful pilot should show that:

1. students can access the lab with minimal technical friction;
2. students understand the mission and rules of engagement;
3. most students can complete or meaningfully attempt the lab;
4. students can use trace evidence in their reports;
5. instructors can grade submissions using the rubric;
6. the lab fits naturally into the course topic;
7. the hosted model saves preparation and troubleshooting time.

13. Limits of the Current Pilot

The initial pilot is intentionally focused. It does not attempt to cover every AI security topic or every item in a comprehensive cybersecurity curriculum.

The first goal is not maximum breadth. The first goal is adoption readiness:

Pilot Principle

A lecturer should be able to assign one AI agent security lab next week with minimal preparation, minimal infrastructure burden, and clear assessment evidence.

14. Next Steps for Interested Instructors

Instructors interested in piloting Agent Failure can begin with a small, low-risk trial.

1. Review the product overview and Lab 1 syllabus mapping.
2. Review the student handout, instructor guide, and rubric.
3. Run Lab 1 once as an instructor or teaching assistant.
4. Assign the lab to a small group or class.
5. Collect student reports and platform traces.
6. Complete the pilot feedback form.
7. Discuss whether the lab should be adopted, revised, or expanded.

15. Contact

Contact Placeholder

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